

EE344: Milestone 3

MON - 10

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Project Description

Auto-transforming Robot

- To build a multi-modal robot which transforms on user's indication.

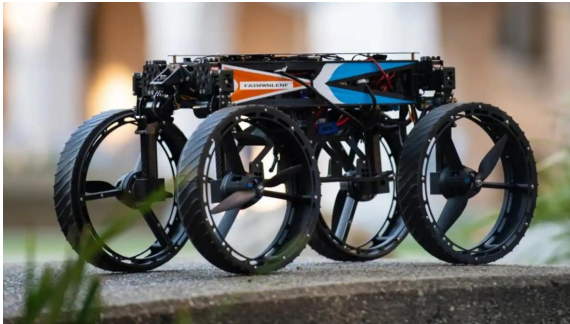


Figure 1: Caltech M4

Feedback Received

Milestone II review:

- Make micro-deadlines to complete subparts before assembly.
- Acquire materials for mechanical parts so that they can be manufactured on short notice.

Problems analysed by us:

- Multiple iterations are required to ensure a well-fitting prototype with clearance
- Need for simpler GUI working with minimal firmware changes in flight controller software in testing.

Addressing the issues:

- We performed three iterations of manufacturing for assembly.
- Switched from Ardupilot to iNAV testing with less-code modification.

Gantt Chart Updates

Layout, CAD & Initial Testing				
Motor integration & testing for all 3 modes	Neel - Sthiti	100%	3/3/25	45724
Communication Setup	Sthiti	100%	3/3/25	3/4/25
PCB Assembly and Testing	Harshil	100%	3/3/25	3/5/25
Finalize Chassis Design	Arin - Saumya	100%	3/3/25	3/10/25
Finalize Tire Assembly Design	Saumya- Arin	100%	3/3/25	3/8/25
First prototype manufacturing	Arin - Saumya - Harshil	100%	3/3/25	3/10/25
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Transformation PoC	Saumya- Arin	100%	3/13/25	3/18/25
PoC Testing - I	Neel - Saumya	100%	3/13/25	3/21/25
Milestone III Evaluation	All	0%	3/22/25	3/27/25

Figure 2: Updated Gantt Chart

Task 1

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PoC Testing - I	Neel - Saumya	100%	3/13/25	3/21/25
Milestone III Evaluation	All	0%	3/22/25	3/27/25

Figure 3: Task 1

Individual Testing of all Modes

- We have individually done testing for all 3 modes - drive, drone, and transformation.
- Drive mode: All 4 DC motors are working by interfacing using the L298N.
- Drone mode: All 4 BLDC motors are working through the SpeedyBee ESC.
- Transformation: All 4 servo motors are rotating as expected to transform from drive to drone.
- All tests were done by sending commands using the RC controller to SpeedyBee.

Task 2

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PoC Testing - I	Neel - Saumya	100%	3/13/25	3/21/25
Milestone III Evaluation	All	0%	3/22/25	3/27/25

Figure 4: Task 2

Changing Flight Controller Firmware

Available Firmware

iNav: We have switched to INAV for testing out the mechanical assembly. INAV-configurator as a GUI allows easy integration of changing the servo outputs for dc motors and servo transformations. Compared to Ardupilot where we had modified the firmware, we only modified the config files in the firmware leading to easy integration.

- We have tested RC connections with INAV
- We have sent command via joystick to test transform and drive ability.
- We are working to suitably modify the code to fit our use case and deal with edge cases.

Task 3

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Milestone III Evaluation	All	0%	3/22/25	3/27/25

Figure 5: Task 3

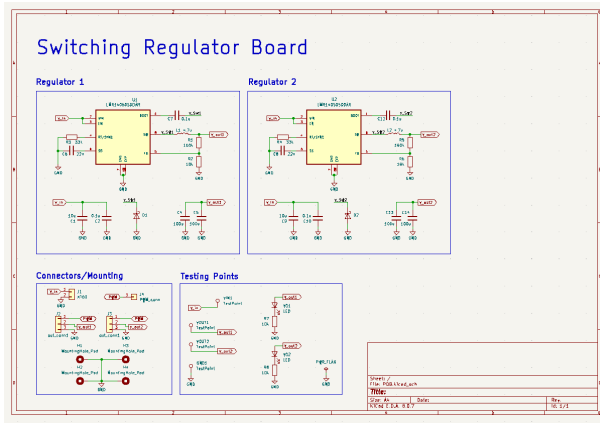


Figure 6: Overall Schematic

PCB Order Form

Audit-trail of PCB fabrication

Project title: PDS - Auto Transforming Robot Tag

Group ID: PCB-V0

Student point of contact (name and roll no.): Adyesh Singh (22.B1266)

Undertaking:

1. I certify that my team has thoroughly reviewed the correctness of schematic and layout of this PCB design.
2. We all understand that all PCBs that we need immediately have been submitted in this order.
3. We all understand that we will not be allowed to submit any more orders up to one week from "Approval for mask printing and start of PCB fabrication" [step 4 in table below]

Signature of all group members: [Signatures]

Justification for cases where all group member signatures are not present above:

Sr. no.	Task	Reviewed/executed by (name & sign)	Date & Time	Remarks
1	Layout footprint and floorplan review (WEL) Expected time to completion: 1 day	Arunadakash/Rajesh	17/01/2025	Note: Students should take photo of components placed on printout of board layout(s) on A4 sheet and show these during review in step 2 below
2	Layout approval and process selection (WEL) Expected time to completion: 1 day	Maheshwar/Ankur	01/02/2025	Recommended PCB format: A. Single-sided, no solder mask B. Double-sided, no solder mask C. With solder mask D. External vendor
3	CAM file review process compatibility (PCB Lab) Expected time to completion: 1 day	Hresh/Srinidhi/Vinush		
4	Approval for mask printing and start of PCB fabrication (PCB Lab) Expected time to completion: 5 days	Yadnik		Order ID:
5	Receipt of PCB (from PCB lab)	Received by: Name: Roll no.:		

Figure 7: Reviewed and Filled

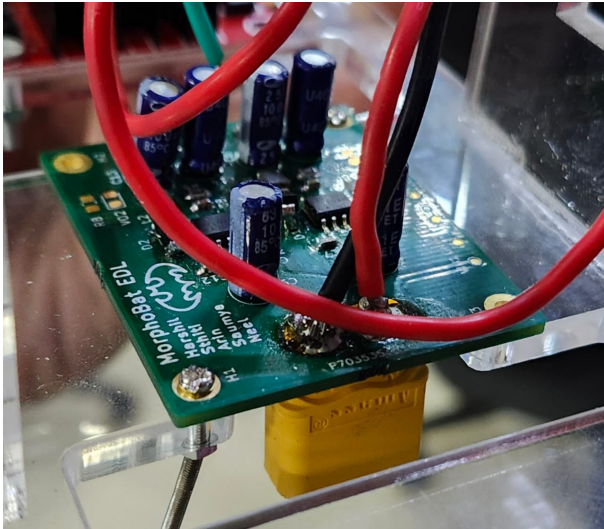


Figure 8: Soldered PCB

- The Power Distribution Board has been designed using the LMR14050 Buck Regulator IC.
- Schematic for one IC and subsequently the overall circuit including 2 Buck IC's, external connections, and test points has been completed.
- PCB layout according to manufacturing constraints has been completed. Board dimensions are 48mm x 40mm.
- The board has been received and tested, it is giving the correct output and driving Servo motor successfully.

Task 4

Layout, CAD & Initial Testing				
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PoC Testing - I	Neel - Saumya	100%	3/13/25	3/21/25
Milestone III Evaluation	All	0%	3/22/25	3/27/25

Figure 9: Task 4

Chassis Assembly Design

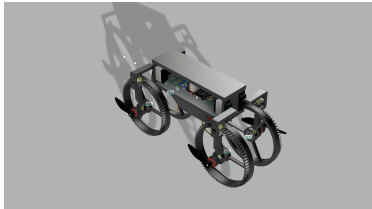


Figure 10: CAD Model-Drive



Figure 11: CAD Model-Drone

Improvements in Design:

- Finger joints used instead of the angle brackets.
- Battery compartment designed to be placed beneath COM. Slotting done to keep battery stable during flight.
- Plates have been added to the 5th and 6th side for stability.

Chassis Assembly



Figure 12: Assembly-first version

Final Assembled Chassis

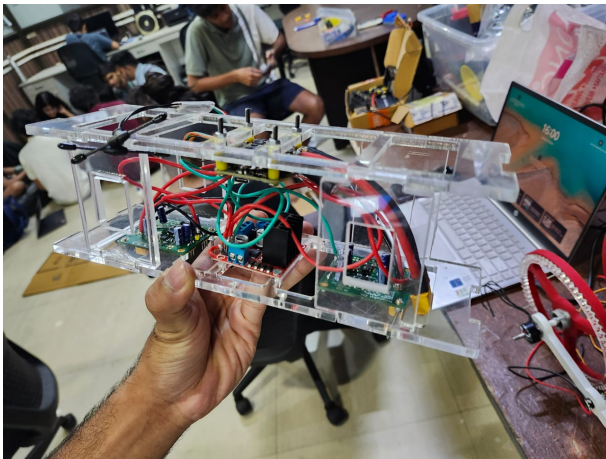


Figure 13: Assembled Chassis

Chassis Assembly

- Side plates and upper plate and lower support have been laser cut. On finding clearance issues for servo motors on side plates, they have been re-manufactured.
- Battery compartment has been manufactured as first iteration. Finger joints are used to join individual parts. (Angle brackets give inaccurate measure, tape was used for rudimentary version)
- The fifth and sixth sides were also added to make the entire chassis stable.
- Motors are integrated and speedybee is fixed on places, final wiring for product finish remains.
- Model chassis dimensions are 25.4 cm x 8 cm x 8cm.

Task 5

Layout, CAD & Initial Testing				
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PoC Testing - I	Neel - Saumya	100%	3/13/25	3/21/25
Milestone III Evaluation	All	0%	3/22/25	3/27/25

Figure 14: Task 5

Tire Assembly

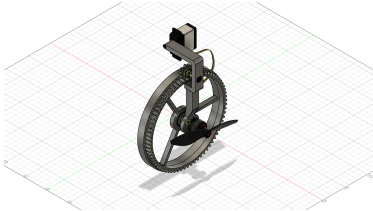


Figure 15: CAD Model of Wheel Assembly



Figure 16: Wheel Assembly - First version

- The video link for tire assembly working driven by DC motor and gear set-up. This is the first version.
- The final version with modified leg design and assembled drive mode can be viewed here.

Tire Assembly

- Design of tire assembly was modified after first prototype to fit servo horn hold and to fix DC motor.
- We manufactured (3D printed) the wheel twice to reduce weight and increase clearance between wheel and chassis.
- The wheel was 3D-printed at 20% infill to reduce weight of the model.
- The leg is designed in without wire compartment in the latest iteration for better integration.
- The leg is attached to servo horn and rotates on command to servo. This a first version transforming video.

Task 6

Layout, CAD & Initial Testing				
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PoC Testing - I	Neel - Saumya	100%	3/13/25	3/21/25
Milestone III Evaluation	All	0%	3/22/25	3/27/25

Figure 17: Task 6

Manufacturing Process

- The parts of the chassis (top plate, bottom plate and side plates) have been laser cut for precision.
- The first iteration of wheels were 3D printed, the next iteration was laser cut.
- The gears were also laser cut for precise spokes.
- The parts of leg assembly were 3D printed.



Figure 18: Laser cutting on acrylic sheet

Task 7

Layout, CAD & Initial Testing				
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PoC Testing - I	Neel - Saumya	100%	3/13/25	3/21/25
Milestone III Evaluation	All	0%	3/22/25	3/27/25

Figure 19: Task 7

Final Assembled PoC

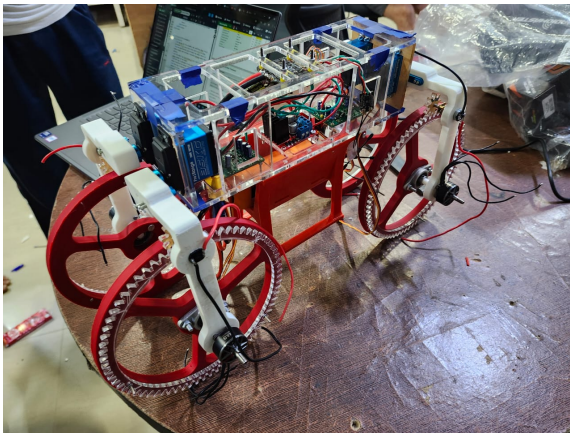


Figure 20: Final Assembled PoC

Task 8

Layout, CAD & Initial Testing				
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PoC Testing - I	Neel - Saumya	100%	3/13/25	3/21/25
Milestone III Evaluation	All	0%	3/22/25	3/27/25

Figure 21: Task 8

- Video shows the prototype of morphobot, commands are sent from RC module and executed - transformation mode.
- The servo motors are given commands from the RC controller and rotate 90 degrees as shown.
- [\[Video Link\]](#)

Task 9 - Demo

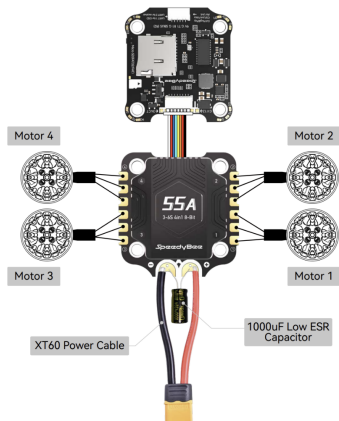
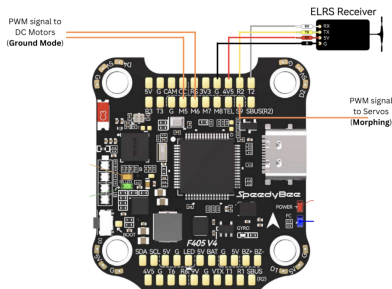
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PoC Testing - I	Neel - Saumya	100%	3/13/25	3/21/25
Milestone III Evaluation	All	0%	3/22/25	3/27/25

Figure 22: Updated Gantt Chart

Proof of Concept

- Video shows the prototype of morphobot, commands are sent from RC module and executed - drive mode.
- Mechanically, we have identified and improved upon tire assembly and design.
- We have tested BLDC motors, drone mode will be tested after model is finalized to avoid risking crashes due to mechanical malfunction.
- [Video Link]

Circuit Diagram: SpeedyBee Connections



Source: SpeedyBee F405V4 Documentation

Figure 23: Connections to SpeedyBee F405 V4.

Left: Front side, Right: Back side

Block Diagram

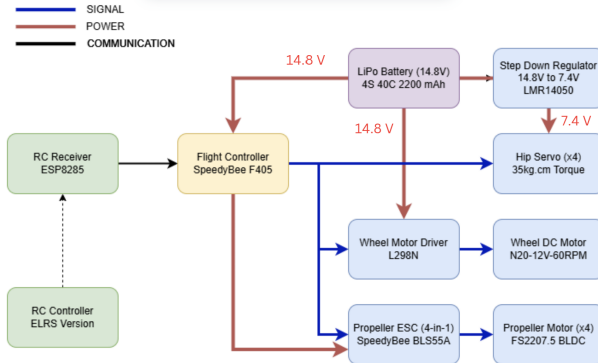


Figure 24: Overall Block Diagram

Work Distribution

Final Testing & Evaluation				
Finalize CAD	Arin - Saumya	0%	3/28/25	3/29/25
Final prototype manufacturing & assembly	Saumya - Harshil	0%	3/30/25	4/1/25
Software implementation on final prototype	Neel - Sthiti	0%	4/2/25	4/5/25
PoC Testing - II	Neel - Saumya	0%	4/2/25	4/5/25
End Report and Final Presentation	All	0%	4/6/25	4/12/25

Figure 25: Work responsibility allocation

Mechanical Assembly:

- After testing Drive and Drone mode in use, suitable end-stage adjustments to be made.
- Final prototype to be manufactured after above design decisions.

Software Integration:

- Smooth control of bot as drone and as rover, and smooth transformation.
- Testing time of operation (time of flight) and seamless communication.

Electrical Final Checkouts:

- All electrical connections have been made and they are working properly.
- Final tests will be run again with greater stress scenarios (such as running on an almost discharged battery).
- Electrical connections will be checked and then isolated and firmly connected using electrical tape.

Project Management Plan for Final Demo

Progress

- All tasks allocated in milestone III have been completed.
- PCB has been soldered and tested.
- Mechanical assembly has been done after multiple iterations.

Management Plan

- Debugging and Testing performed in at least groups of two.
- Ideally, one person from software and one from mechanical to be present.
- Each member aware of all design decisions and working.

All progress has been documented in OneNote and separate markers for subsystems.

Thank you!